

Ahmed Khalid Humoudi

Robotics and Automation Engineer

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🌐 www.linkedin.com/in/ahmed-humoudi-904127317 🔗 portfolio-omega-ten-g6ozj4wxdx.vercel.app

PROFILE

Dedicated robotics engineer with hands-on experience in control, automation, AI integration, and system development. Passionate and seeking opportunities to apply technical expertise to real-world applications.

Portfolio: portfolio-omega-ten-g6ozj4wxdx.vercel.app - contains detailed project information, visualization, and implementation.

EDUCATION

German International University (GIU), Bachelor Degree, 2026

09/2022 – 07/2026 | Cairo, Egypt

Major: Robotics and Automation

GPA: 0.87

Ranked among the top 3 students in the Robotics Major.

Global International School (Al-Kon), High School Degree, 2022

09/2019 – 06/2022 | Jeddah, Saudi Arabia

IGCSE Score: 120%

PROFESSIONAL EXPERIENCE

Automation Engineering Intern, Mactech Automation Solutions

07/2026 – 08/2026 | Jeddah, Saudi Arabia

Developed PLC programs using Siemens TIA Portal and PLCSIM for S7-1500 PLCs and ET 200SP I/O systems. Worked with LAD, Function Blocks, digital/analog I/O, PID control, HMI development, and PROFINET communication.

Mechatronics Engineer Intern, Hiryo

02/2026 – 07/2026 | Cairo, Egypt

Designed and developed a Bluetooth-based alarm and tracking device with real-time alert and location features, integrating embedded hardware, firmware, and mobile connectivity to deliver a reliable end-to-end system.

Physics Lab Assistant, German International University

02/2024 – 06/2024 | Cairo, Egypt

Conducted hands-on laboratory supervision and technical guidance, assisting in experimental procedures and the practical application of theoretical concepts.

PROJECTS

Bachelor Thesis

Reinforcement Learning-Based Path Planning for Warehouse Robots in Dynamic Environments:

Developed an RL-based navigation system for autonomous warehouse robots operating in dynamic environments with obstacle avoidance and adaptive path planning.

Multi-Robot Task Formation: Centralized ROS2 system, obstacle avoidance, path planning, formation control.

PLC-Based Stepper Motor Motion Control: controlling stepper motor speed, direction, position control.

UR5E Pick and Place: UR5e robot integrated with ROS 2, MoveIt 2, and RViz for trajectory planning.

Oven Control System: ATmega32, temperature monitoring, fan control, and safety lock system.

Ball and Beam Control System: PID-based ball stabilization on a beam using Arduino Uno.

4-DOF Robotic Arm: Arduino-based 4-DOF arm with inverse kinematics and trajectory planning.

TECHNICAL SKILLS

Industrial Automation: Siemens TIA Portal, PLCSIM, WinCC HMI, PROFINET, LAD, Function Blocks, PID Control

Robotics: ROS 2, MoveIt 2, RViz 2, Gazebo, MATLAB, Simulink

Programming: C, C++, Python, JavaScript

Embedded Systems: Arduino, Raspberry Pi, ATmega32, ESP32

Artificial Intelligence: Machine Learning, Deep Learning, Reinforcement Learning

Languages: Arabic (Native), English (C1), German (A2)

COURSES

From Wire to PLC: A Bootcamp in Industrial Automation, *Udemy* — Online

PLC Programming, *HA Consulting Group* — In Person

Modern Robotics: Foundations of Robot Motion, *Northwestern University, Coursera* — Online

Introduction to Artificial Intelligence, *IBM, Coursera* — Online

MATLAB, Simulink, Reinforcement Learning Onramps, *MATLAB Self-Paced Courses* — Online